

Unsupervised Learning and Clustering

1: Unsupervised learning is a part of machine learning which works differently from supervised because there is no teacher (supervisor) involved to guide the machine. In this approach the machine is given with data that has **no labels or categories**.

Clustering is used to group similar data points together. Clustering algorithms work by repeatedly moving data points closer to the center of their group (cluster) and farther from points in other groups.

The difference between regression, classification and clustering are:

Classification assigns data to predefined categories, Regression predicts continuous numerical values, and Clustering groups unlabeled data based on natural similarities.

Real life Example:

supervised learning:

Credit card fraud detection (Classification): is a crucial application of machine learning in the financial sector.

Unsupervised Learning:

Customer segmentation: This involves the identification of distinct levels and patterns within market segmentation, allowing companies to better understand their customer base.

2: Clustering Algorithms:

Hierarchical Clustering is an unsupervised learning algorithm that builds clusters in a hierarchical (tree-like) structure.

♦ Advantages of Hierarchical Clustering

- No need to choose K initially
- Easy to visualize using dendrogram
- Shows hierarchy of clusters
- Useful for understanding data structure

♦ Limitations

- Computationally expensive (slow for large data)
- Once merged, clusters cannot be undone (in agglomerative)
- Sensitive to noise and outliers

K-Means is an unsupervised machine learning algorithm used to group similar data points together.

♦ Advantages of K-Means

- Simple to understand
- Easy to implement
- Fast for large datasets
- Works well for numeric data

◆ Limitations of K-Means

- Must choose K manually
- Sensitive to outliers
- Not good for non-spherical clusters

Density Based Spatial Clustering of Applications with Noise: is an unsupervised clustering algorithm that groups points based on density instead of distance to centroids.

◆ Advantages of DBSCAN

- No need to choose K
- Detects outliers automatically
- Handles irregular shapes
- Works well when clusters vary in shape

◆ Limitations of DBSCAN

- Choosing eps is tricky
- Struggles when cluster densities vary
- Not ideal for very high-dimensional data

3:Clustering Metrics:

- Elbow Method(SSE): measures the sum of squared errors or inertia within cluster across different values of k.
- The Silhouette Score is a way to measure how good the clusters are in a dataset. It helps us understand how well the data points have been grouped. The score ranges from -1 to 1.
- The Davies-Bouldin Index (DBI) helps us measure how good the clustering is in a dataset. It looks at how tight each cluster is (compactness), and how far apart the clusters are (separation).

Metrics	measures	what a good value looks like
Elbow Methods(SSE)	internal cluster tightness	look for the inflection point where adding more clusters
Silhouette Score	balance between cluster cohesion tightness and separation	closer to +1 indicates well-defined, isolated , and highly distinct clusters
Davies-Bouldin index	ratio of intra-cluster scatter to intra-cluster separation	closer to () is optimal, indicating low internal scatter and high distance between clusters

4: Choosing k and interpreting segments:

- The number of clusters (k) is usually chosen using the Elbow Method. This method plots the Sum of Squared Errors (SSE) for different values of k. The best value is where the curve forms an "elbow," indicating that adding more clusters provides only a small improvement. The Silhouette Score can also be used to measure how well-separated the clusters are.
- In the wholesale distributor project, what does it mean when a cluster has high Fresh + Milk spend vs high Grocery + Detergents Paper spend? A cluster with high Fresh and Milk spending represents customers such as restaurants, hotels, or cafés that frequently purchase fresh food and dairy products. A cluster with high Grocery and Detergents Paper spending represents supermarkets or retail stores that buy packaged groceries and cleaning products in large quantities.
- We exclude Channel and Region because they are categorical variables, while K-Means works best with numerical features. The goal is to group customers based on their purchasing behavior, not based on their location or sales channel. These variables can still be used later to help interpret the resulting clusters.

5: Real-World Case Study: Customer Segmentation Using K-Means Clustering

Goal: The goal of the project was to segment supermarket customers into different groups based on their purchasing behavior. This helps the business create targeted marketing campaigns and improve customer satisfaction.

Data Used: The project used customer purchase data, including annual spending on products such as Fresh, Milk, Grocery, Frozen, Detergents Paper, and Delicatessen.

Clustering Method Applied: The project applied K-Means Clustering to group customers with similar spending patterns. The Elbow Method was used to determine the optimal number of clusters.

Key Results and Insights: The model identified several customer segments with different buying habits. For example, one group spent more on fresh products, while another spent more on grocery and cleaning products. These insights helped the business personalize promotions, improve inventory planning, and better understand customer needs.